INCH-POUND
MIL-C-55302/118D
7 June 1993
SUPERSEDING
MIL-C-55302/118C
31 October 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, DECADE INCREMENT 20 THROUGH 110 AND 128 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302.

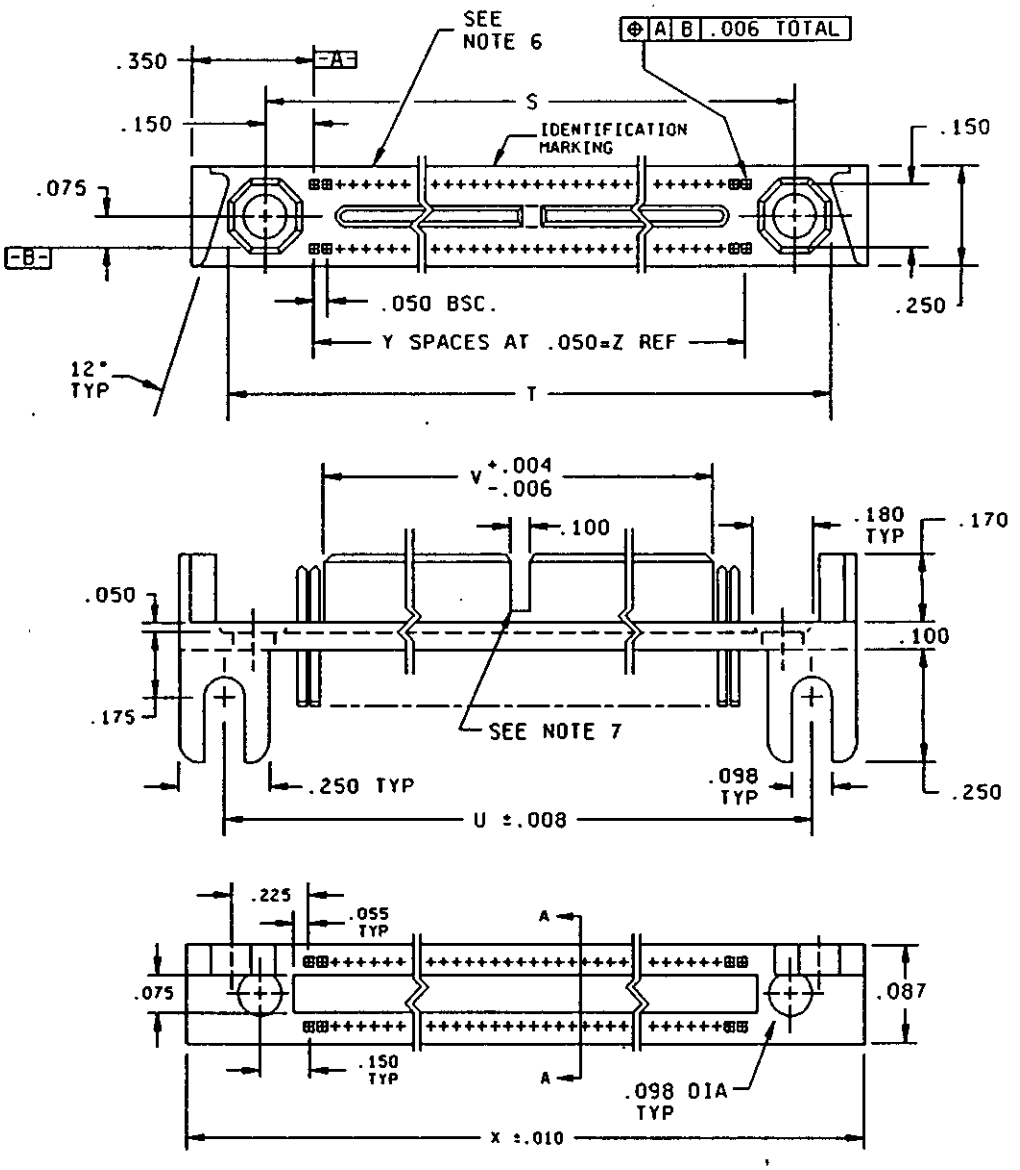
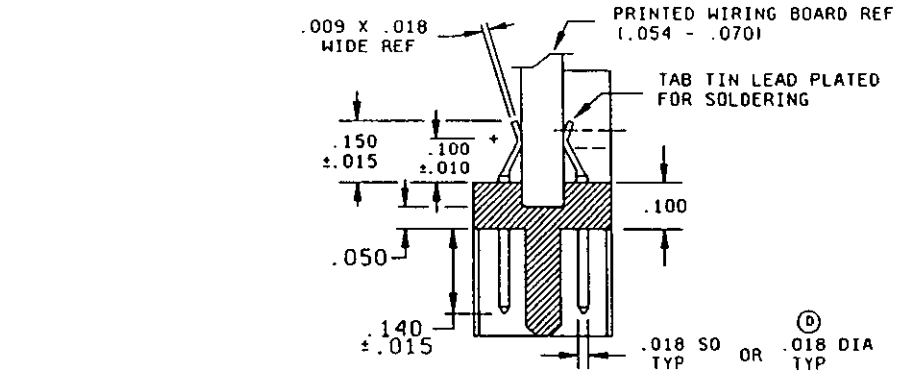


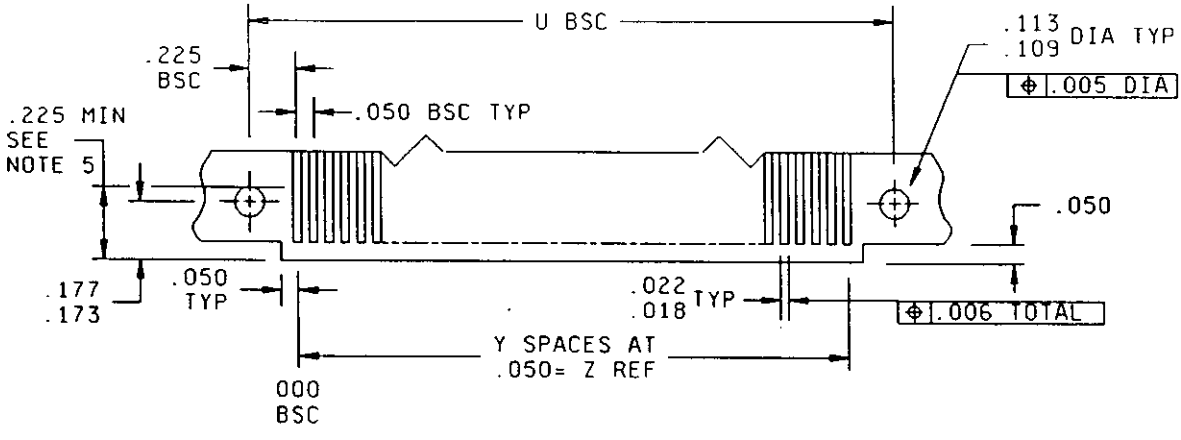
FIGURE 1. Connectors, plug without pin protectors, .050 (1.27 mm) spacing for style 1.

(D) denotes changes.

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SECTION A-A



Inches	mm	Inches	mm	Inches	mm
.004	0.10	.054	1.37	.150	3.81
.005	0.13	.055	1.40	.170	4.32
.006	0.15	.070	1.78	.173	4.39
.008	0.20	.075	1.90	.175	4.44
.009	0.23	.087	2.21	.177	4.50
.010	0.25	.098	2.49	.180	4.57
.015	0.38	.100	2.54	.225	5.72
.018	0.46	.109	2.77	.250	6.35
.022	0.56	.113	2.87	.350	8.89
.050	1.27	.140	3.56		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/119(USAF).
5. Minimum pad length on printed wiring board.
6. Numbers indicating end positions stamped this side only.
7. Except 20 through 50 position.

FIGURE 1. Connectors, plug without pin protectors, .050 (1.27 mm) spacing for style 1 - Continued.

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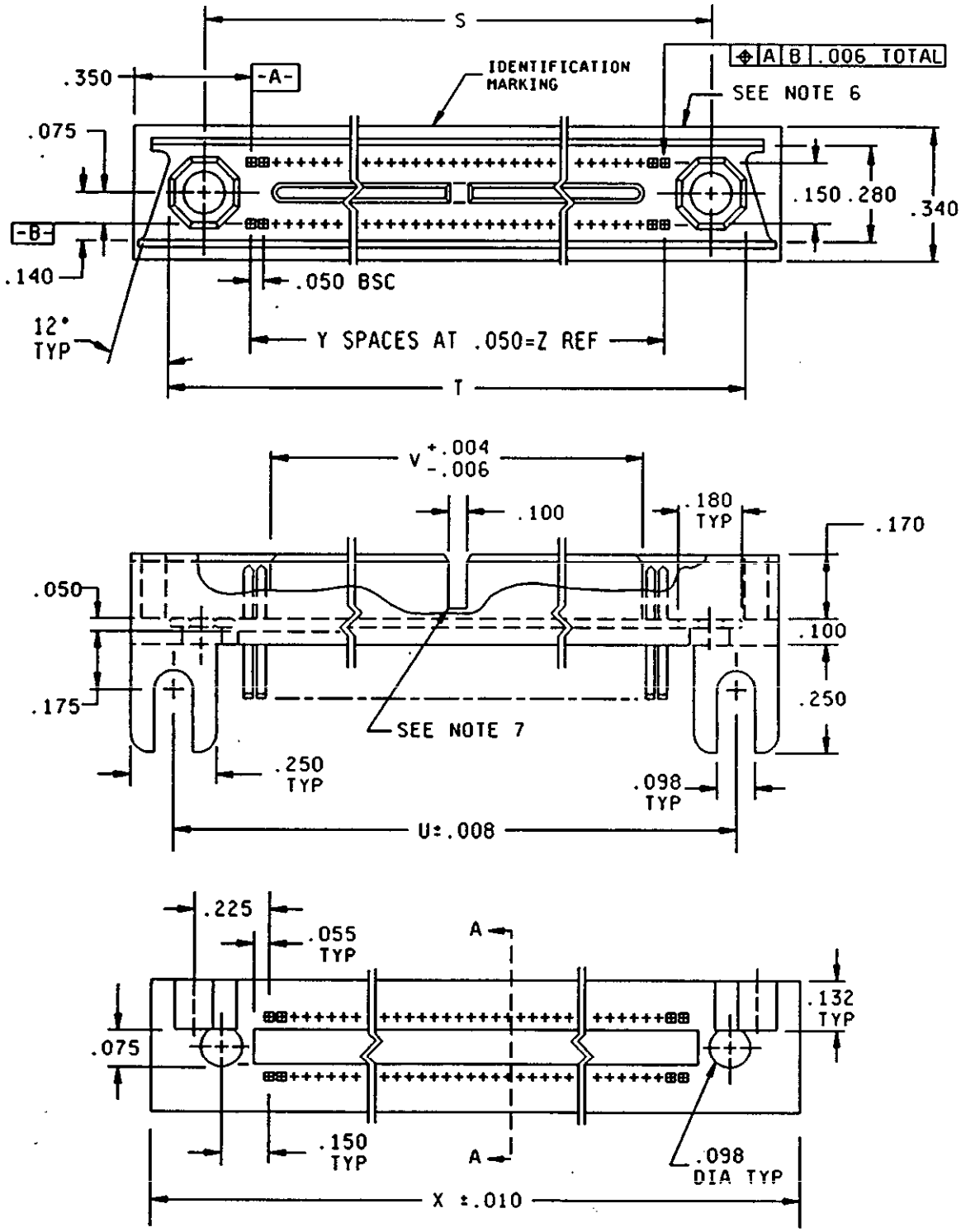
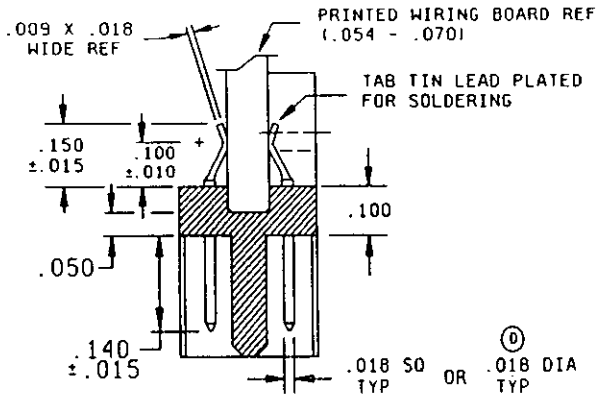
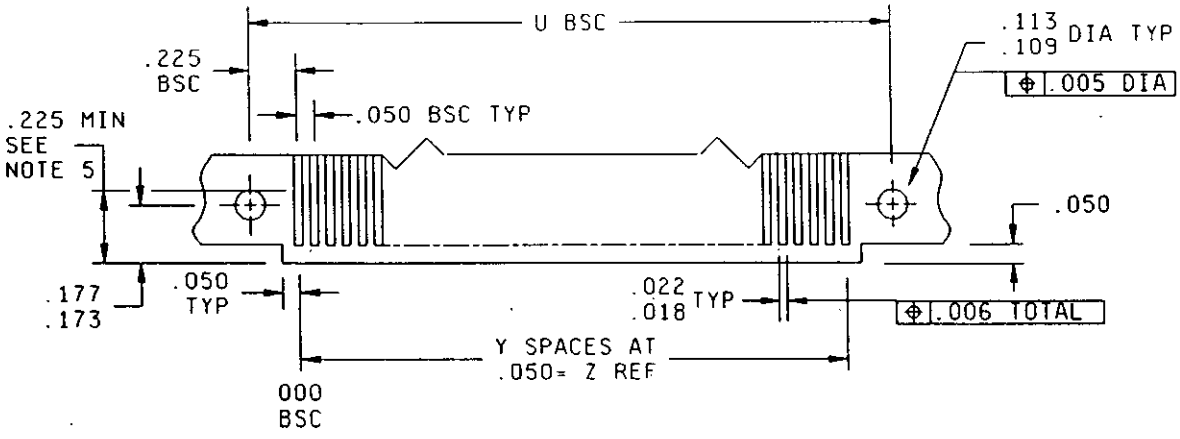


FIGURE 2. Connectors, plug with pin protectors, .050 (1.27 mm) spacing for style 2.

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SECTION A-A



Inches	mm	Inches	mm	Inches	mm
.004	0.10	.054	1.37	.170	4.32
.005	0.13	.055	1.40	.173	4.39
.006	0.15	.070	1.78	.175	4.44
.008	0.20	.075	1.90	.177	4.50
.009	0.23	.098	2.49	.180	4.57
.010	0.25	.100	2.54	.225	5.72
.015	0.38	.109	2.77	.250	6.35
.018	0.46	.113	2.87	.280	7.11
.022	0.56	.140	3.56	.340	8.64
.050	1.27	.150	3.81	.350	8.89

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/119(USAF).
5. Minimum pad length on printed wiring board.
6. Numbers indicating end positions stamped this side only.
7. Except 20 through 50 position.

FIGURE 2. Connectors, plug with pin protectors, .050 (1.27 mm) spacing for style 2 - Continued.

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TABLE I. Dash numbers and dimensions for style 1.

Dash number	Number of contacts	Dimensions							Contact ident. numbers
		S	T	U	V	X	Y	Z	
01	20	.750 (19.05)	.970 (24.64)	.900 (22.86)	.290 (7.37)	1.150 (29.21)	9	.450 (11.43)	1-10
02	30	1.000 (25.40)	1.220 (30.99)	1.150 (29.21)	.540 (13.72)	1.400 (35.56)	14	.700 (17.78)	1-15
03	40	1.250 (31.75)	1.470 (37.34)	1.400 (35.56)	.790 (20.07)	1.650 (41.91)	19	.950 (24.13)	1-20
04	50	1.500 (38.10)	1.720 (43.69)	1.650 (26.42)	1.040 (26.42)	1.900 (48.26)	24	1.200 (30.48)	1-25
05	60	1.750 (44.45)	1.970 (50.04)	1.900 (48.26)	1.290 (32.77)	2.150 (54.61)	29	1.450 (36.83)	1-30
06	70	2.000 (50.80)	2.220 (56.39)	2.150 (54.61)	1.540 (39.12)	2.400 (60.96)	34	1.700 (43.18)	1-35
07	80	2.250 (57.15)	2.470 (62.74)	2.400 (60.96)	1.790 (45.47)	2.650 (67.31)	39	1.950 (49.53)	1-40
08	90	2.500 (63.50)	2.720 (69.09)	2.650 (67.31)	2.040 (51.82)	2.900 (73.66)	44	2.200 (55.88)	1-45
09	100	2.750 (69.85)	2.970 (75.44)	2.900 (73.66)	2.290 (58.17)	3.150 (80.01)	49	2.450 (62.23)	1-50
10	110	3.000 (76.20)	3.220 (81.79)	3.150 (80.01)	2.540 (64.52)	3.400 (86.36)	54	2.700 (68.58)	1-55
11	128	3.450 (87.63)	3.670 (93.22)	3.600 (91.44)	2.990 (75.95)	3.850 (97.79)	63	3.150 (80.01)	1-64

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TABLE II. Dash numbers and dimensions for style 2.

Dash number	Number of contacts	Dimensions							Contact ident. numbers
		S	T	U	V	X	Y	Z	
12	20	.750 (19.05)	.970 (24.64)	.900 (22.86)	.290 (7.37)	1.150 (29.21)	9	.450 (11.43)	1-10
12	30	1.000 (25.40)	1.220 (30.99)	1.150 (29.21)	.540 (13.72)	1.400 (35.56)	14	.700 (17.78)	1-15
14	40	1.250 (31.75)	1.470 (37.34)	1.400 (35.56)	.790 (20.07)	1.650 (41.91)	19	.950 (24.13)	1-20
15	50	1.500 (38.10)	1.720 (43.69)	1.650 (26.42)	1.040 (26.42)	1.900 (48.26)	24	1.200 (30.48)	1-25
16	60	1.750 (44.45)	1.970 (50.04)	1.900 (48.26)	1.290 (32.77)	2.150 (54.61)	29	1.450 (36.83)	1-30
17	70	2.000 (50.80)	2.220 (56.39)	2.150 (54.61)	1.540 (39.12)	2.400 (60.96)	34	1.700 (43.18)	1-35
18	80	2.250 (57.15)	2.470 (62.74)	2.400 (60.96)	1.790 (45.47)	2.650 (67.31)	39	1.950 (49.53)	1-40
19	90	2.500 (63.50)	2.720 (69.09)	2.650 (67.31)	2.040 (51.82)	2.900 (73.66)	44	2.200 (55.88)	1-45
20	100	2.750 (69.85)	2.970 (75.44)	2.900 (73.66)	2.290 (58.17)	3.150 (80.01)	49	2.450 (62.23)	1-50
21	110	3.000 (76.20)	3.220 (81.79)	3.150 (80.01)	2.540 (64.52)	3.400 (86.36)	54	2.700 (68.58)	1-55
22	128	3.450 (87.63)	3.670 (93.22)	3.600 (91.44)	2.990 (75.95)	3.850 (97.79)	63	3.150 (80.01)	1-64

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1 and 2.

Material:

(D) Insulator body: Insulator material shall be in accordance with MIL-C-55302 or MIL-M-24519 type GLCP-30F.

(D) Contact material: Contact material shall be in accordance with MIL-C-55302 or beryllium copper alloy CDA174 in accordance with ASTM B768.

(D) Plating: Contact plating shall be gold over nickel in accordance with MIL-C-55302 in the contact engagement area for .100 inch minimum length. The termination area plating shall be tin lead over nickel in accordance with MIL-C-55302 for .100 inch minimum length. The remainder of the contact shall be plated with nickel in accordance with MIL-C-55302.

Contact identification: See figures 1 and 2.

Contact rating: 1.5 amperes per contact, maximum.

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Operating temperature: -65°C to 125°C.

Keying: See MIL-C-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .30 and the withdrawal force in pounds shall be a minimum of .03 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: .75 pound, minimum.

Dielectric withstanding voltage:

Sea level: 600 volts rms.

High altitude: 150 volts rms.

Solderability: Each connector shall be subjected to method 208 of MIL-STD-202 (solder termination only).

Part number: M55302/118- (dash number from tables I and II).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3907-04)

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